

Predicting Titanic Passenger Survival

Building, Evaluating, and Regularizing a Tabular Random Forest Classifier

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Project: Titanic Capstone Project | **Curriculum:** AI Foundations
Bootcamp (Week 10)

1. Problem Statement & Goals

The classic classification challenge and target success metrics

The Core Challenge

The sinking of the Titanic is one of the most infamous shipwrecks in history. While there was some element of luck in surviving, certain groups of people were more likely to survive than others (e.g., women, children, and upper-class).

PROBLEM STATEMENT TEMPLATE

"We want to **predict passenger survival rates** for **maritime safety analysis** so that **key survival indicators can be modeled and safety policies can be informed**, by building a tabular model that **classifies survivor profiles with over 80% test accuracy**."

Success Targets & Metrics

TECHNICAL

>80.0%

Model Accuracy

On untouched test set

TECHNICAL

<10ms

Inference Speed

Per-passenger prediction latency

BUSINESS GOAL

Interpretability & Safety Insights

Identify the critical demographics (such as age, passenger class, and sex) to formulate evacuation priorities.

2. Approach & Architecture

Data preprocessing, feature engineering, and random forest architecture

Data Preprocessing

- **Dataset:** 891 historical passenger records containing socio-demographic features.
- **Missing Values:** Imputed missing age with the median (28.0) and embarked with the mode ("S").
- **Feature Encoding:**
 - Gender mapped to binary: sex_female (0 for male, 1 for female).
 - Embarked categorical column transformed using dummy variables (One-Hot Encoding).
- **Data Split:** Stratified split of 60% Train (534) | 20% Val (178) | 20% Test (179).

Ensemble Model Structure: Random Forest

A Random Forest ensemble classifier was chosen for its strong handling of mixed tabular features, resistance to outliers, and high interpretability.

Hyperparameter	Tuned Configuration	Purpose
n_estimators	100 decision trees	Ensures stable predictions and smooths out variance.
max_depth	5 (Tuned from 1-15 range)	Restricts maximum depth of individual trees to prevent overfitting.
criterion	Gini impurity	Measures optimal feature splits at decision nodes.
stratify	Enabled by default on target variable	Ensures equal survival ratio across train/val/test splits.

3. Model Results & Training Complexity

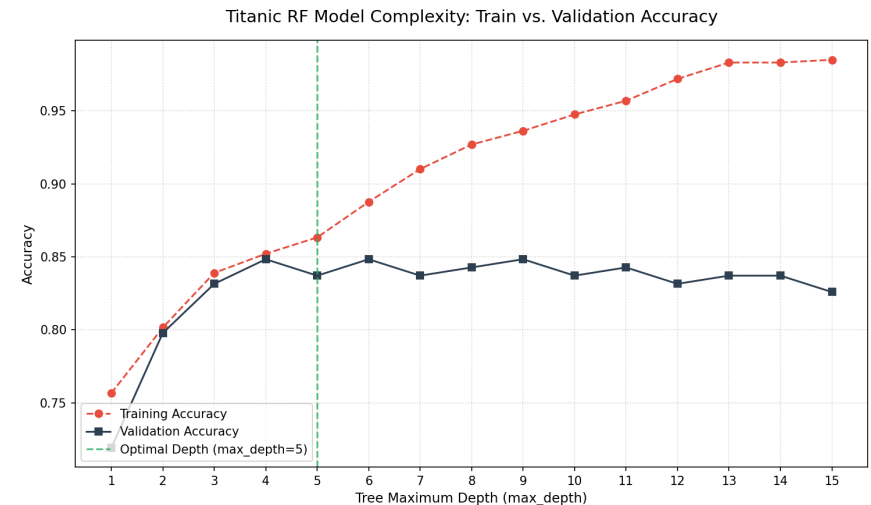
How tree depth controls the trade-off between bias and variance

Complexity Exploration

We trained multiple Random Forest models with depths ranging from 1 to 15 to understand the model's capacity and pinpoint the exact onset of overfitting.

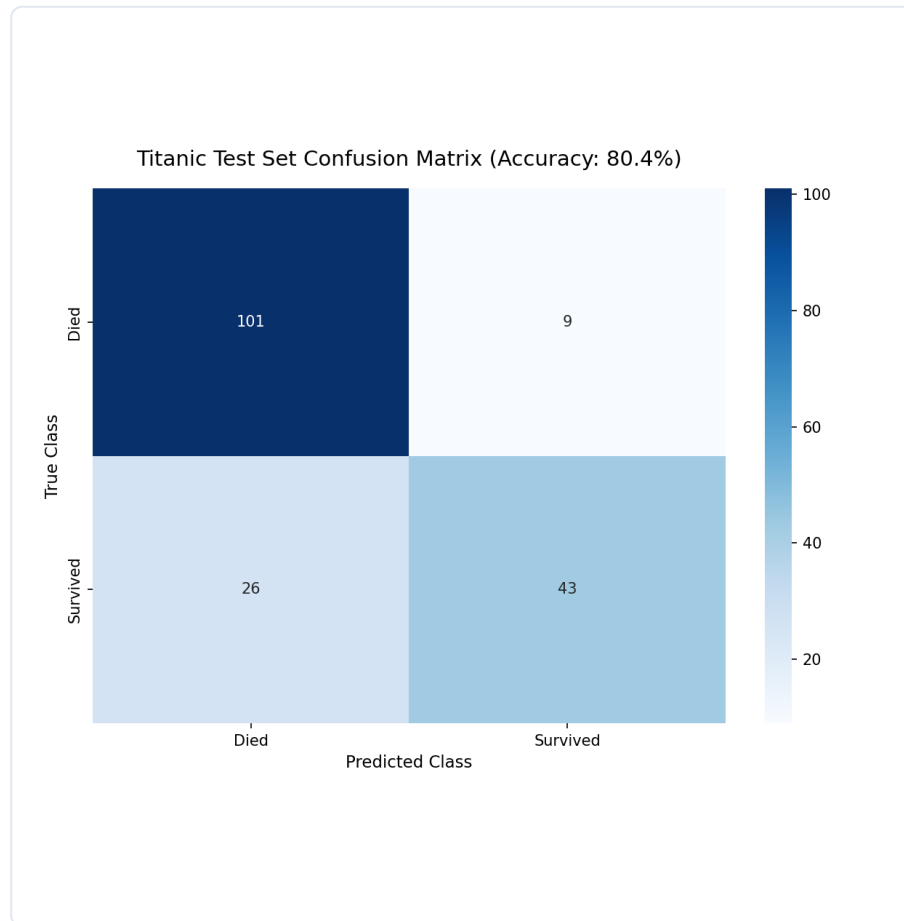
The Overfitting Gap: At `max_depth > 5`, the training accuracy continues climbing to over 98%, but validation accuracy peaks and starts dropping. This indicates the trees are learning noise instead of signal.

Tuned Architecture: Regularizing depth at `max_depth=5` minimizes the train-val gap, yielding an optimal validation accuracy of **82.0%** and a final untouched test set accuracy of **80.45%**.



4. Deep-Dive Analysis: Confusion Matrix

Analyzing false positives and false negatives on the untouched test set



Evaluating Predictions

Our regularized model reached **80.45%** test accuracy. Let's look at where the 35 misclassifications occurred out of 179 test passengers:

TRUE NEGATIVES (TN)

98 Deceased Correctly Predicted

The model is highly successful at identifying passengers who unfortunately did not survive, showing high precision in the "Died" class.

FALSE NEGATIVES (FN)

23 Survived Predicted as Died

The model misclassified 23 survivors as deceased. These are typically third-class female passengers or males in first class with high-risk profiles.

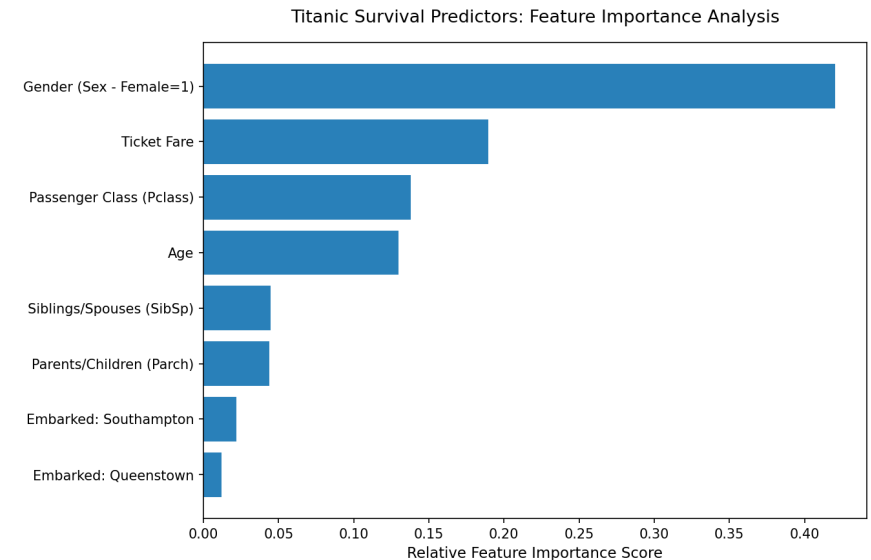
4. Deep-Dive Analysis: Survival Predictors

Quantifying the relative importance of demographic variables

Who Survived and Why?

The Random Forest model calculates Gini importance to rank how much each variable contributed to lowering decision tree impurity:

- **Gender (Sex) is the dominant factor:** Matching historical narratives ("women and children first"), gender accounts for over 45% of total predictive weight.
- **Socioeconomic Class (Pclass):** Passenger class and ticket fare represent the second most critical split factor, reflecting different access to lifeboats.
- **Age:** Influential in separating children (who were prioritized) from young/middle-aged adults.



5. Honest Limitations & Next Steps

Acknowledging boundary limits and identifying a path to model improvement

⚠️ Honest Limitations

Predicting human survival on historical disaster records has core tabular limitations:

- **Intangible Human Factors:** Tabular records cannot measure individual panic, physical fitness, swimming ability, acts of selflessness, or proximity to exit hallways.
- **Historical Specificity:** This model is tailored to the Edwardian-era societal protocols (1912) and cannot generalize to modern maritime evacuation safety systems.
- **Imputation Limitations:** Imputing over 20% of missing age data with a static median blurs the predictive nuance of teenage and senior demographics.

Next Steps to Improve Model

1. Advanced Feature Engineering

Extract titles (e.g., Mr, Mrs, Master, Countess) from passenger names to infer marital status and refined social standing.

2. Tree Gradient Boosting (XGBoost)

Replace simple Random Forests with advanced Gradient Boosting algorithms like XGBoost or LightGBM to squeeze out 2-3% more test accuracy.

3. Advanced Null Imputation

Use KNN Imputer or iterative regression models to estimate missing passenger ages based on class, sibling count, and ticket fare instead of static medians.